

Map Guided Lab (Part 1)

Robert Thomas Kohn

10/27/25

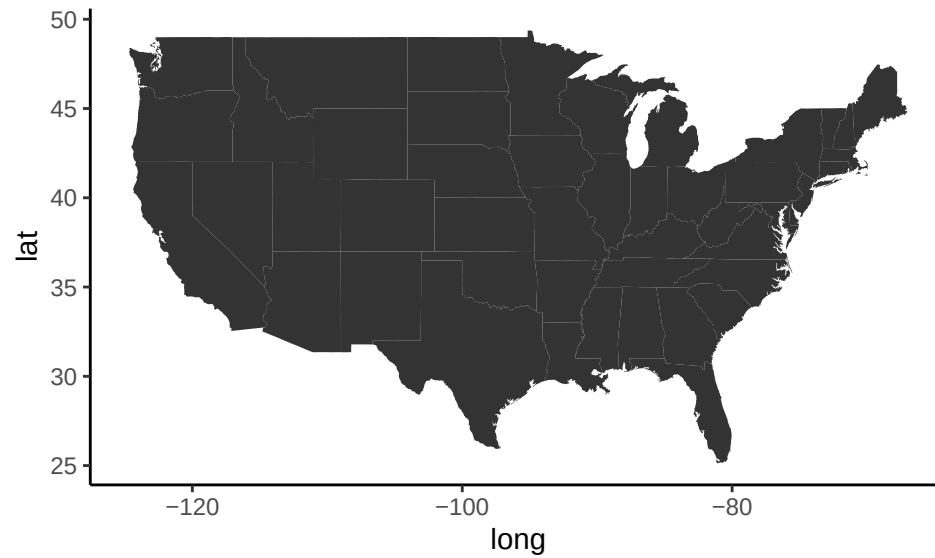
Large Figures

```
#Graph 1
require( maps )

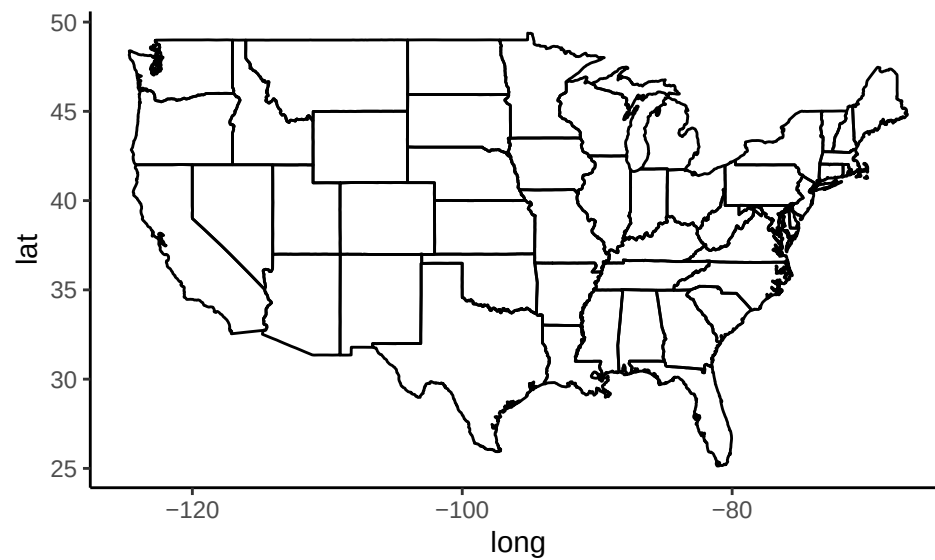
## Loading required package: maps
## Warning: package 'maps' was built under R version 4.4.3
##
## Attaching package: 'maps'
## The following object is masked from 'package:purrr':
##
##      map
require(mapproj)

## Loading required package: mapproj
## Warning: package 'mapproj' was built under R version 4.4.3
states = map_data( "state" )
head( states )

##           long      lat group order  region subregion
## 1 -87.46201 30.38968     1     1 alabama    <NA>
## 2 -87.48493 30.37249     1     2 alabama    <NA>
## 3 -87.52503 30.37249     1     3 alabama    <NA>
## 4 -87.53076 30.33239     1     4 alabama    <NA>
## 5 -87.57087 30.32665     1     5 alabama    <NA>
## 6 -87.58806 30.32665     1     6 alabama    <NA>
gf_polygon( lat ~ long , data = states , group = ~ group )
```



```
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black " , size = 0.5 )
```



```
theme_set(theme_map())
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black " , size = 0.5 )
```



```
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black " , size = 0.5 )%>
```

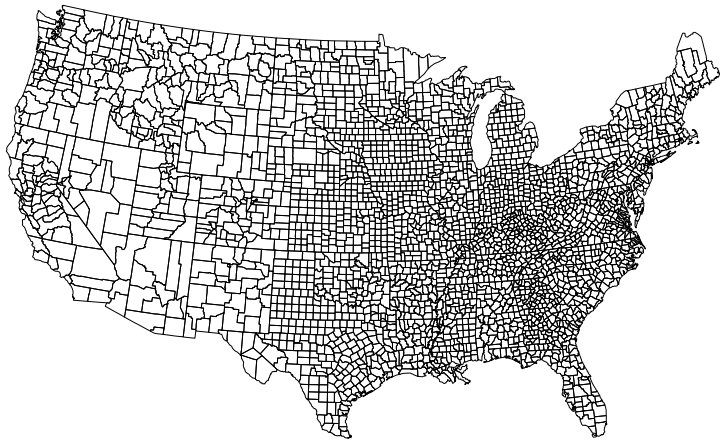


#Graph 2

```
county = map_data( "county" )
head( county )
```

```
##      long      lat group order  region subregion
## 1 -86.50517 32.34920     1     1  alabama  autauga
## 2 -86.53382 32.35493     1     2  alabama  autauga
## 3 -86.54527 32.36639     1     3  alabama  autauga
## 4 -86.55673 32.37785     1     4  alabama  autauga
## 5 -86.57966 32.38357     1     5  alabama  autauga
## 6 -86.59111 32.37785     1     6  alabama  autauga
```

```
gf_polygon( lat ~ long , data = county , group = ~ group , fill = NA , color = " black " , size = 0.1 )%>
```

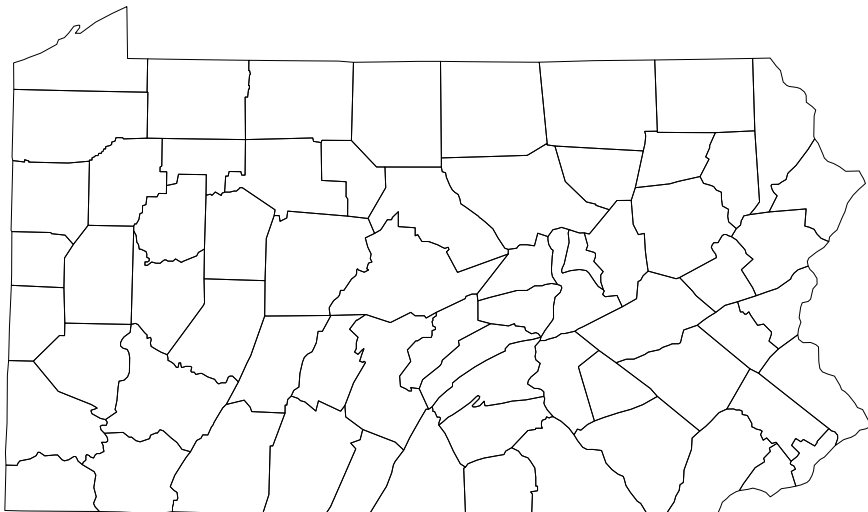


#Graph 3

```
penn = map_data( "county" , "pennsylvania" )
head(penn)
```

```
##      long      lat group order      region subregion
## 1 -77.44670 39.96954     1      1 pennsylvania      adams
## 2 -77.42952 39.98672     1      2 pennsylvania      adams
## 3 -77.37222 40.00391     1      3 pennsylvania      adams
## 4 -77.32065 40.01537     1      4 pennsylvania      adams
## 5 -77.23471 40.02683     1      5 pennsylvania      adams
## 6 -77.18887 40.03256     1      6 pennsylvania      adams
```

```
gf_polygon( lat ~ long , data = penn , group = ~ group , fill = NA , color = " black " , size = 0.1 )%>%
```



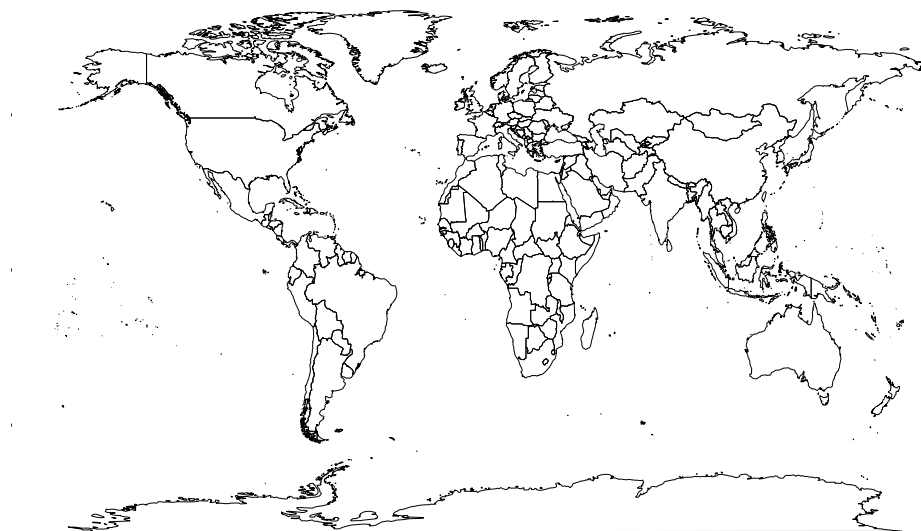
World Map Outline

#Graph 4

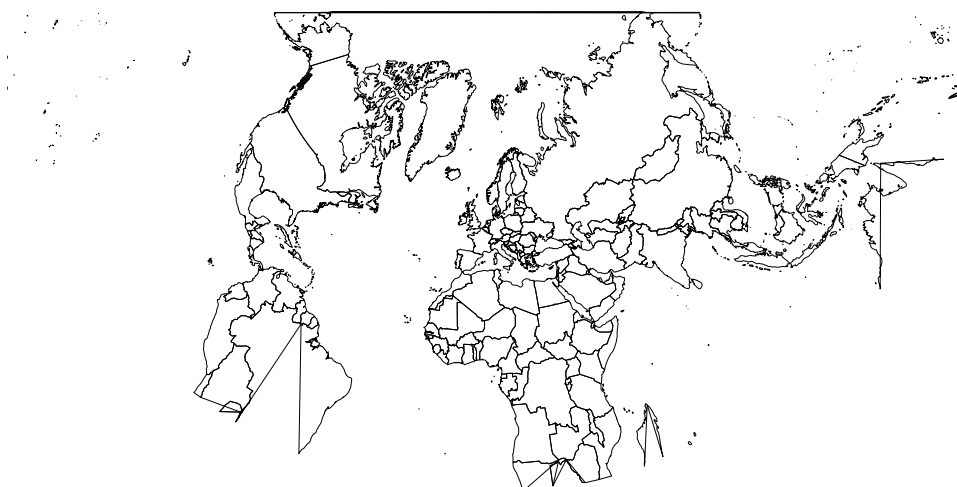
```
world = map_data("world")
head(world)
```

```
##      long      lat group order region subregion
## 1 -69.89912 12.45200     1     1  Aruba      <NA>
## 2 -69.89571 12.42300     1     2  Aruba      <NA>
## 3 -69.94219 12.43853     1     3  Aruba      <NA>
## 4 -70.00415 12.50049     1     4  Aruba      <NA>
## 5 -70.06612 12.54697     1     5  Aruba      <NA>
## 6 -70.05088 12.59707     1     6  Aruba      <NA>
```

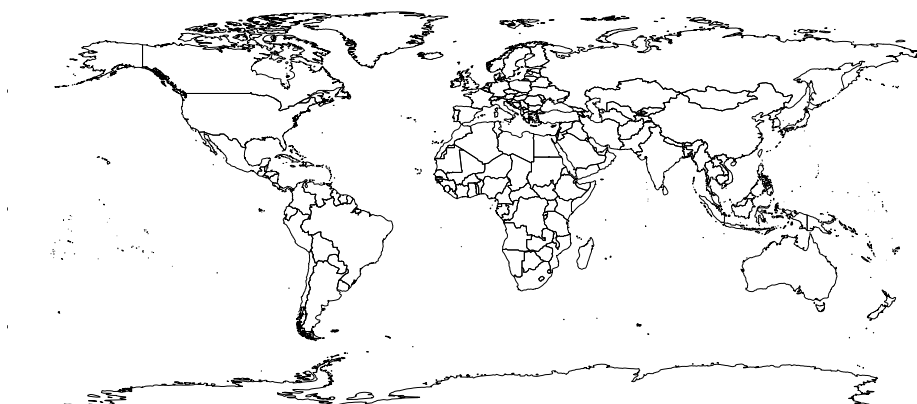
```
gf_polygon( lat ~ long , data = world , group = ~ group , fill = NA , color = " black " , size = 0.1 )
```



```
gf_polygon( lat ~ long , data = world , group = ~ group , fill = NA , color = " black " , size = 0.1 )%>
```



```
gf_polygon( lat ~ long , data = world , group = ~ group , fill = NA , color = " black " , size = 0.1 )%>
```



#Graph 5

```
japan = map_data("world", "japan")
head(japan)
```

##	long	lat	group	order	region	subregion
## 1	123.8887	24.28013	1	1	Japan	Iriomote Jima
## 2	123.8256	24.26606	1	2	Japan	Iriomote Jima
## 3	123.7498	24.28330	1	3	Japan	Iriomote Jima
## 4	123.6807	24.28804	1	4	Japan	Iriomote Jima
## 5	123.6798	24.31777	1	5	Japan	Iriomote Jima
## 6	123.7523	24.34849	1	6	Japan	Iriomote Jima

```
gf_polygon( lat ~ long , data = japan , group = ~ group , fill = NA , color = " black " , size = 0.1 )%>
```



Dot Overlay (Using Cities)

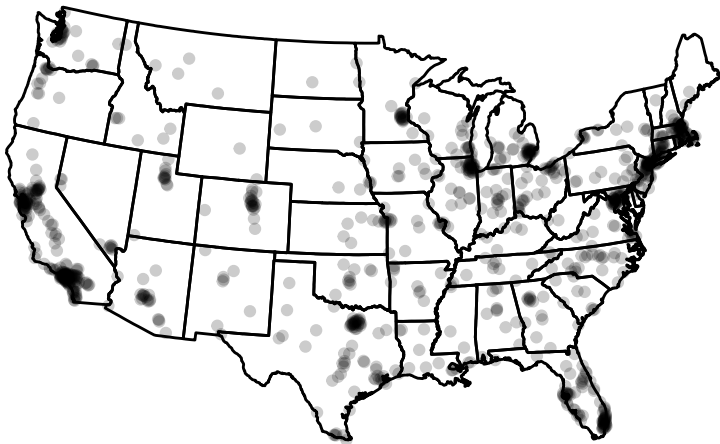
```
data( "us.cities" )
head( us.cities )
```

##	name	country.etc	pop	lat	long	capital
## 1	Abilene TX	TX	113888	32.45	-99.74	0
## 2	Akron OH	OH	206634	41.08	-81.52	0
## 3	Alameda CA	CA	70069	37.77	-122.26	0
## 4	Albany GA	GA	75510	31.58	-84.18	0
## 5	Albany NY	NY	93576	42.67	-73.80	2
## 6	Albany OR	OR	45535	44.62	-123.09	0

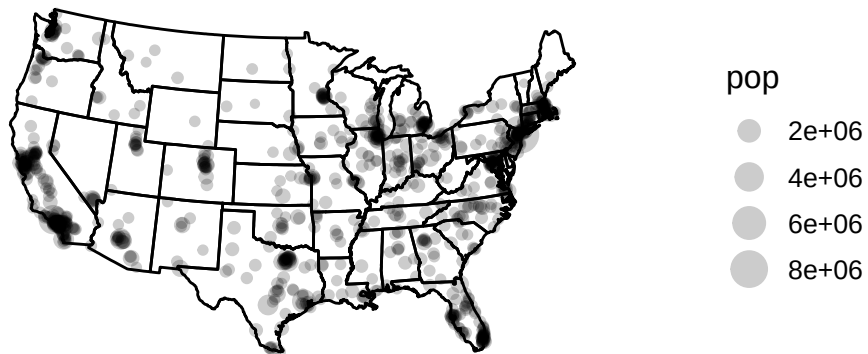
```
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black " , size = 0.5 )%>
```



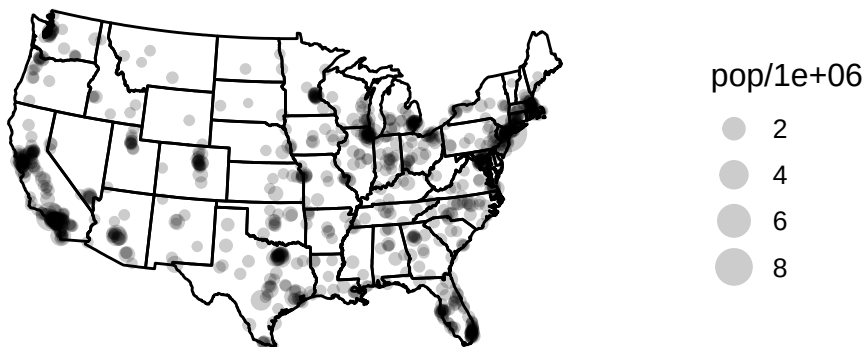
```
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black ", size = 0.5 )%>
```



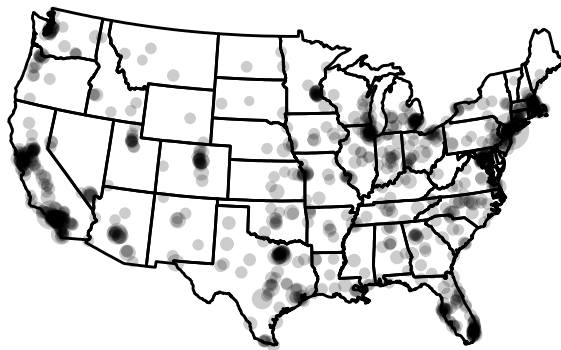
```
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black ", size = 0.5 )%>
```



```
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black " , size = 0.5 )%
```



```
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black " , size = 0.5 )%
```



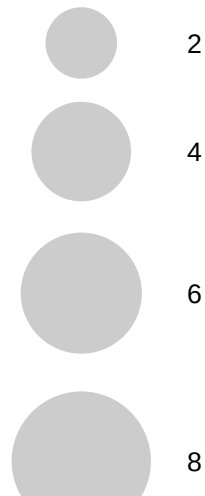
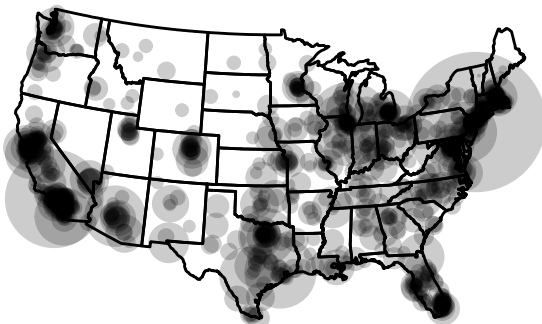
Population
(Millions)



```
gf_polygon( lat ~ long , data = states , group = ~ group , fill = NA , color = " black " , size = 0.5 ) %>
```

Population
(Millions)

Location and Pop Size in USA Cities



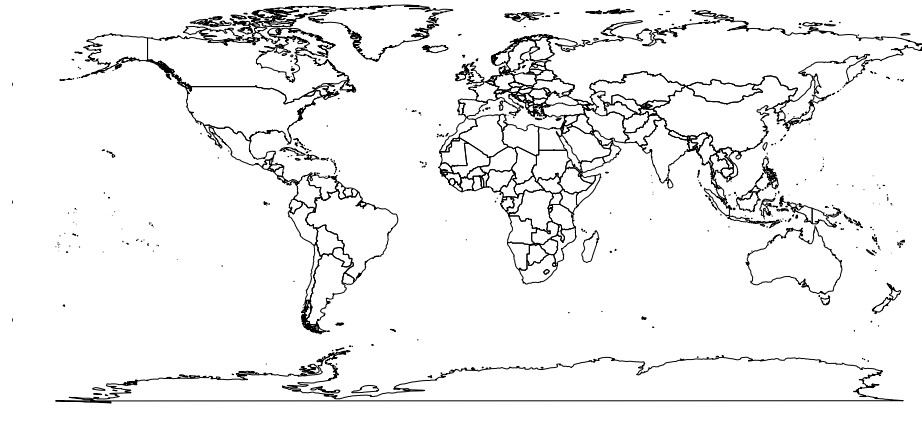
#Graph 7

```
data(world.cities)
head(world.cities)
```

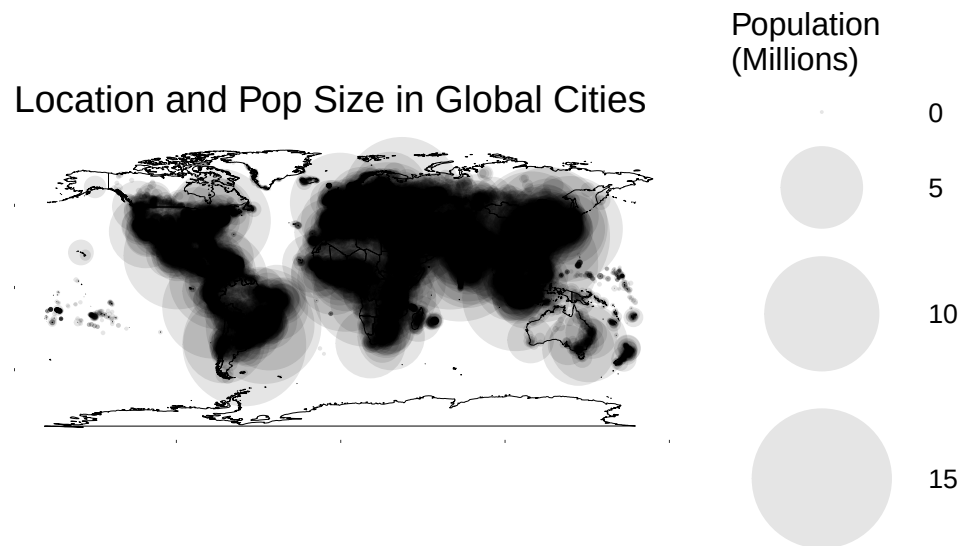
```
##          name country.etc  pop  lat  long capital
## 1 'Abasan al-Jadidah  Palestine  5629 31.31 34.34      0
## 2 'Abasan al-Kabirah  Palestine 18999 31.32 34.35      0
## 3      'Abdul Hakim    Pakistan 47788 30.55 72.11      0
## 4 'Abdullah-as-Salam    Kuwait 21817 29.36 47.98      0
## 5      'Abud          Palestine  2456 32.03 35.07      0
## 6      'Abwein          Palestine  3434 32.03 35.20      0
```

```
gf_polygon( lat ~ long , data = world , group = ~ group , fill = NA , color = " black " , size = 0.05 ) %>
```

Location and Pop Size in Global Cities

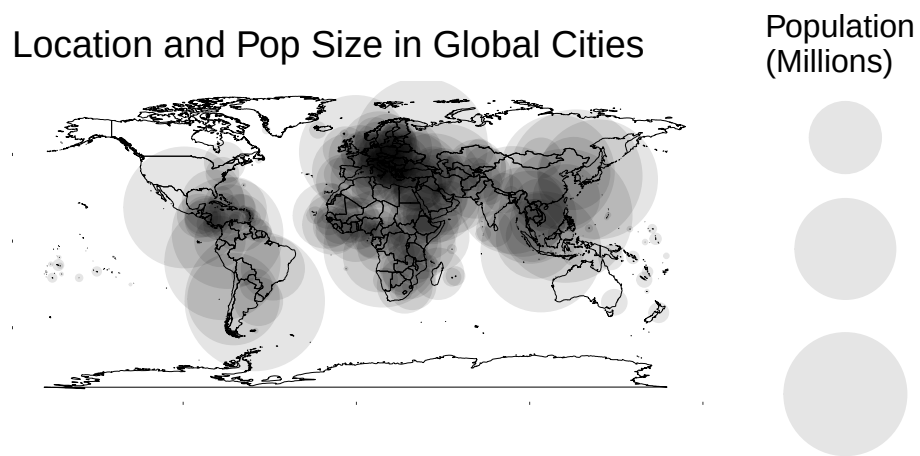


```
gf_polygon( lat ~ long , data = world , group = ~ group , fill = NA , color = " black " , size = 0.05 ) %>
```



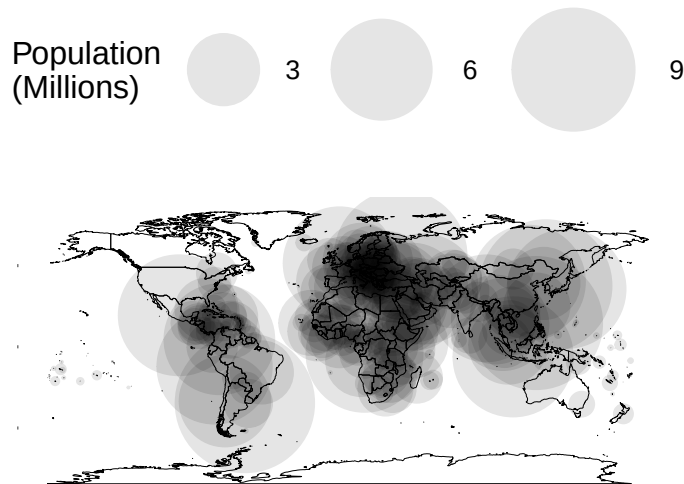
```
gf_polygon( lat ~ long , data = world , group = ~ group , fill = NA , color = " black " , size = 0.05 ) %>
```

Location and Pop Size in Global Cities



```
gf_polygon( lat ~ long , data = world , group = ~ group , fill = NA , color = " black " , size = 0.05 ) %>
```

Location and Pop Size in Global Cities

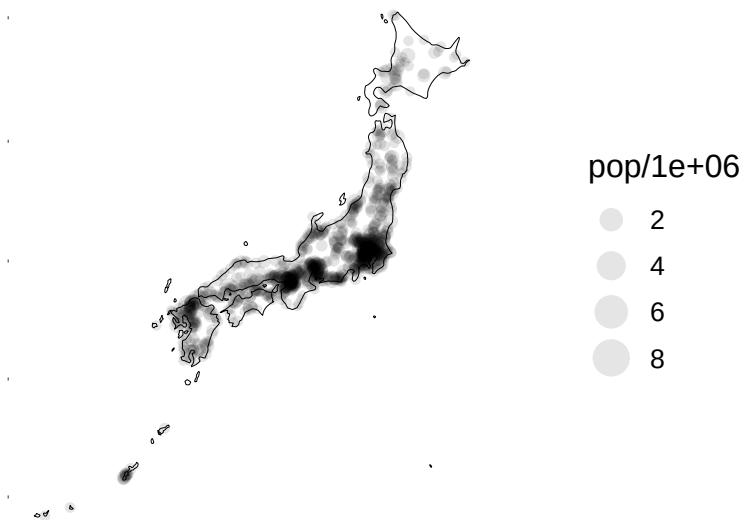


#Graph 8

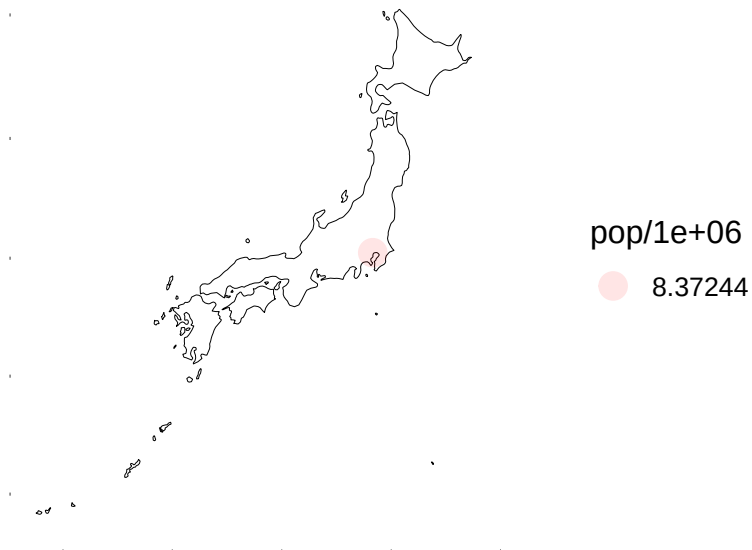
```
gf_polygon( lat ~ long , data = japan , group = ~ group , fill = NA , color = " black " , size = 0.1 ) %>
```



```
gf_polygon( lat ~ long , data = japan , group = ~ group , fill = NA , color = " black " , size = 0.1 )%>
```



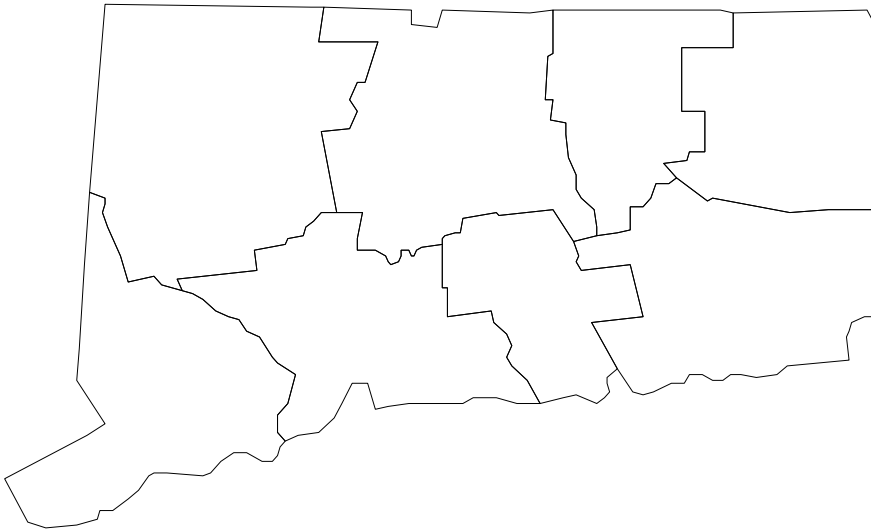
```
gf_polygon( lat ~ long , data = japan , group = ~ group , fill = NA , color = " black " , size = 0.1 )%>
```



Labels on All Cities within State

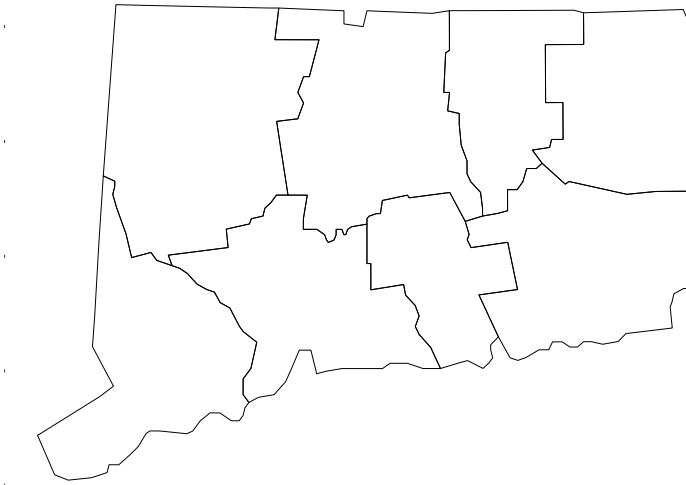
Graph 9

```
ct = map_data( "county", "connecticut" )
gf_polygon( lat~long, data = ct, group = ~group, fill = NA, color = "black", size = 0.05 )
```



```
gf_polygon( lat ~ long , data = ct , group = ~ group , fill = NA , color = " black ", size = 0.05 )%>%
```

Location and Pop Size in Global Cities



```
head(us.cities)
```

##	name	country.etc	pop	lat	long	capital
## 1	Abilene TX	TX	113888	32.45	-99.74	0
## 2	Akron OH	OH	206634	41.08	-81.52	0
## 3	Alameda CA	CA	70069	37.77	-122.26	0
## 4	Albany GA	GA	75510	31.58	-84.18	0
## 5	Albany NY	NY	93576	42.67	-73.80	2
## 6	Albany OR	OR	45535	44.62	-123.09	0

```
gf_point( lat~long, data=subset( us.cities, country.etc == "CT"),alpha = 0.2, size = ~pop, show.legend =
```

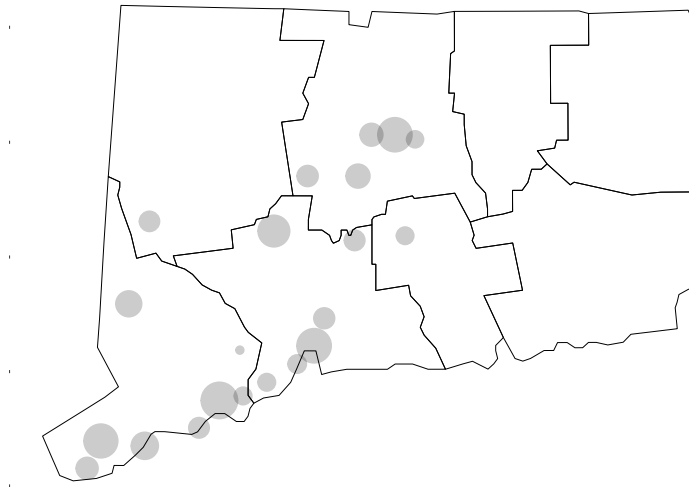


```
#gf_polygon(lat ~ long, data = ct, group = ~group, fill = NA, color = "black", size = 0.05) + gf_point(
library(ggformula)
library(ggplot2)
ggplot() + geom_polygon(data = ct, aes(x = long, y = lat, group = group), fill = NA, color = "black", s
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
```

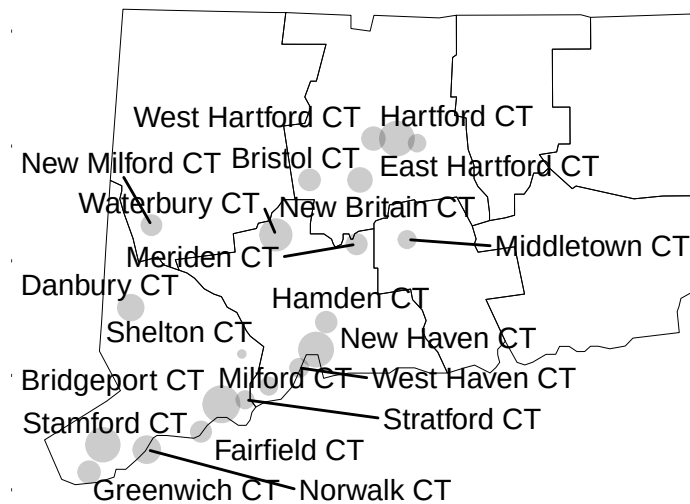
```
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Location and Pop Size in CT



```
ggplot() + geom_polygon(data = ct, aes(x = long, y = lat, group = group), fill = NA, color = "black", s
```

Location and Pop Size in CT



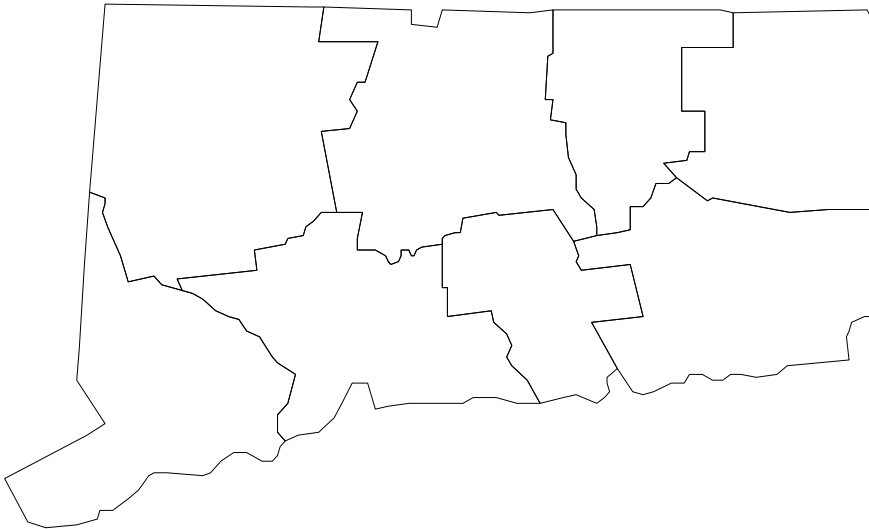
```
ggplot() + geom_polygon(data = ct, aes(x = long, y = lat, group = group), fill = NA, color = "black", s
```

Location and Pop Size in CT



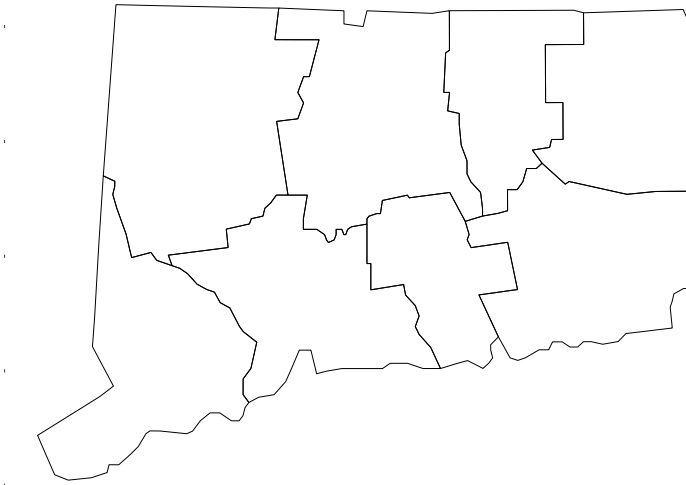
#Graph 10

```
ct = map_data( "county", "connecticut" )
gf_polygon( lat~long, data = ct, group = ~group, fill = NA, color = "black", size = 0.05 )
```



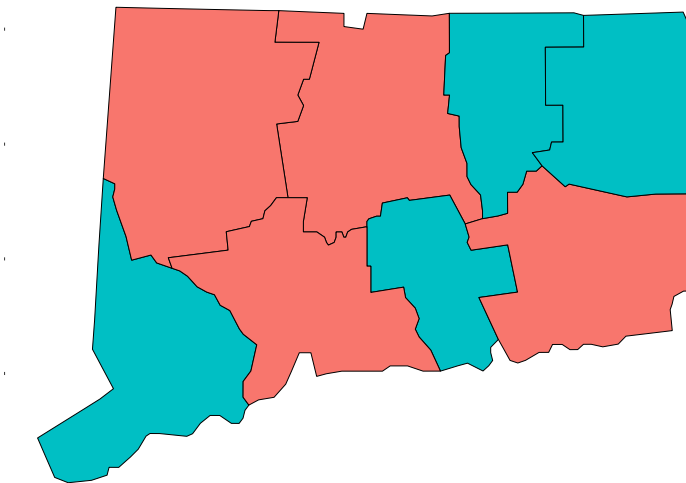
```
gf_polygon( lat~long, data = ct, group = ~group, fill = NA, color = "black", size = 0.05 )>% gf_refine
```

Location and Pop Size in CT Counties



```
gf_polygon( lat ~ long, data = ct , group = ~group,
fill = ~( subregion %in% c("tolland", "middlesex", "fairfield", "windham")),
color = "black", size = 0.05, show.legend = FALSE)%>% gf_refine(coord_map( "conic" , lat0 = 30 ) )%>% g
```

Location and Pop Size in CT Counties



```
centroid.test = ct %>% # Note, you want to use your state dataset (I called mine penn)
group_by(group) %>% # Create averages for each county (group)
mutate(mean.lat = mean(lat), # Calculate the mean latitude & create a mean.lat variable
mean.long = mean(long)) %>% # Calculate the mean longitude & create a mean.long variable
dplyr::select(subregion, mean.lat, mean.long, group) %>% # Keep only these variables
unique() # Get rid of all the duplicate rows we now have

#centroid.test = ct %>% group_by(group) %>% mutate(mean.lat = mean(lat), mean.long = mean(long)) %>% dp

ggplot() + geom_polygon(data = ct, aes(x = long, y = lat, group = group), fill = NA, color = "black",
```

Location and Pop Size in CT

